

ARYAN BHAPKAR

Machine Learning & Robotics Engineer | AI Engineer | Autonomous Systems Researcher

623-272-6544 • abhapk@asu.edu • LinkedIn • GitHub

EDUCATION

Master of Science (M.S.) in Robotics and Autonomous Systems (Artificial Intelligence) Aug 2024 - Dec 2025
Arizona State University, Tempe, AZ

BTech in Artificial Intelligence Aug 2020 - May 2024
NMIMS Mukesh Patel School of Technology Management & Engineering, Mumbai, India

TECHNICAL SKILLS

Machine Learning and AI: Deep Learning, Reinforcement Learning (TD3), CNNs, GANs, LLMs, Fine-Tuning, MLOps, NLP, Model Training, Data Preprocessing, Inference Optimization, Model Evaluation

Robotics & Control: ROS2, MuJoCo, URDF, Robot Kinematics, Trajectory Planning, Kalman Filtering, Dynamic Movement Primitives (DMPs), Inverse Kinematics (IK)

Computer Vision: OpenCV, YOLOv9, Optical Flow, Feature Extraction, Object Detection, Image Preprocessing, Visual Tracking

Programming: Python, SQL, Arduino C, R, Bash

Frameworks & Tools: NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, TensorFlow, AWS (EC2, Lambda, S3, DynamoDB), Matplotlib, Linux, MATLAB, ROS2 Gazebo

EXPERIENCE

Interactive Robotics Laboratory, Tempe, AZ: Research Aide (Advisors: Prof. Heni Ben Amor & Mohamed Mistiri)
Aug 2025 - present

- Developed a MuJoCo-based UR5e simulation for adaptive tabletop manipulation using reinforcement learning and trajectory planning.
- Built an LLM-driven robotic control system with Google Gemini Flash 2.5 to convert natural-language commands into real-time robot actions.

ACADEMIC PROJECTS

StyleGAN-Based Style Mapper Fall 2023

- Designed a StyleGAN-based style transfer system using GAN inversion and JoJoGAN to generate high-resolution images with single-reference style control.

5-DOF Robotic Arm Table Tennis Prototype Spring 2025

- Built a sim-to-real 5-DOF robotic arm in Gazebo/ROS2 using YOLOv9, TD3, OpenCV, Kalman filtering, and optical flow, achieving 95% real-world accuracy and 30ms latency.

RESEARCH PAPERS

Open Set Masked Face Identification System (IEEE ICCVMI) Spring 2023

- Developed an open-set masked face identification system using computer vision and machine learning techniques to improve identity recognition under face-covering conditions. (DOI: 10.1109/CVMI59935.2023.10464743)

EarSiamNet: Leveraging Siamese Networks & SVM for Ear Biometric Authentication (IEEE MoSICOM) Fall 2023

- Developed EarSiamNet, a Siamese-network-based ear biometric authentication model with SVM classification, achieving 97.86% accuracy using HOG feature extraction. (DOI: 10.1109/MoSiCom59118.2023.10458801)

Color Sorting Pick Up and Drop Robotic Arm (IEEE I2CT2) Spring 2024

- Engineered a color-sorting pick-and-place robotic arm using sensor-based object detection for automated sorting in industrial applications. (DOI: 10.1109/I2CT61223.2024.10544011)

Mitigating the Effect of Shadows in Vision-Based Industrial Robot (IEEE ICETA) Fall 2024

- Developed a vision-based industrial robotics pipeline with lighting optimization, image preprocessing, and PatMax for shadow-resilient detection on the KUKA KR4 AGILUS. (DOI: 10.1109/MPSecICETA64837.2025.11118444)